symcoder Phase-3 simplicial demo
standalone simplicial encoder, shadow event #0

1 About this document

This document demonstrates the Phase-3 simplicial-complex one-shot encoder running on its own, with Phase 1 and Phase 2 disabled. The simplicial encoder is deterministically faithful on the full rep alignment table, so it is a valid standalone encoder; this document shows exactly what it consumes (the rep atoms and the alignment table T) and what it emits.

2 Setup

Particle types: electrons = $\{a, b, c\}$, muons = $\{p, q\}$
Group G: $G = S_{\text{electrons}} \times S_{\text{muons}} = S_3 \times S_2$, order = 12

Operations:
$|\mathbf{x}|$ (rank 1, symmetric)
$\mathbf{x} \cdot \mathbf{y}$ (rank 2, symmetric)
$\varepsilon(\mathbf{x}, \mathbf{y}, \mathbf{z})$ (rank 3, antisymmetric)

Event vectors (3-D):
a: $\mathbf{a} = (-1.00,\ -1.00,\ -1.00)$

b: $\mathbf{b} = (-1.00,\ -1.00,\ 0.00)$

c: $\mathbf{c} = (1.00,\ -1.00,\ -1.00)$

p: $\mathbf{p} = (0.00,\ 1.00,\ 0.00)$

q: $\mathbf{q} = (0.00,\ -1.00,\ 0.00)$

3 Phase 3 Encoding (one-shot whole-table; rep size 25, —G—=12)

rep atoms (25): the unpruned set used by Phase 3 — every distinct atom across all G-orbits, not just one canonical representative per orbit.
[**0**]: [0] $|\mathbf{a}|$

[**1**]: [1] $|\mathbf{b}|$

[**2**]: [2] $|\mathbf{c}|$

[**3**]: [3] $|\mathbf{p}|$

[**4**]: [4] $|\mathbf{q}|$

[**5**]: [5] $\mathbf{a} \cdot \mathbf{b}$

[**6**]: [6] $\mathbf{a} \cdot \mathbf{c}$

[**7**]: [7] $\mathbf{b} \cdot \mathbf{c}$

[**8**]: [8] $\mathbf{a} \cdot \mathbf{p}$

[**9**]: [9] $\mathbf{a} \cdot \mathbf{q}$

[**10**]: [10] $\mathbf{b} \cdot \mathbf{p}$

[**11**]: [11] $\mathbf{b} \cdot \mathbf{q}$

[**12**]: [12] $\mathbf{c} \cdot \mathbf{p}$

[**13**]: [13] $\mathbf{c} \cdot \mathbf{q}$

[**14**]: [14] $\mathbf{p} \cdot \mathbf{q}$

[**15**]: [15] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{c})$

[**16**]: [16] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{p})$

[**17**]: [17] $\varepsilon(\mathbf{a}, \mathbf{b}, \mathbf{q})$

[**18**]: [18] $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{p})$

[**19**]: [19] $\varepsilon(\mathbf{a}, \mathbf{c}, \mathbf{q})$

[**20**]: [20] $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{p})$

[**21**]: [21] $\varepsilon(\mathbf{b}, \mathbf{c}, \mathbf{q})$

[**22**]: [22] $\varepsilon(\mathbf{a}, \mathbf{p}, \mathbf{q})$

[**23**]: [23] $\varepsilon(\mathbf{b}, \mathbf{p}, \mathbf{q})$

[**24**]: [24] $\varepsilon(\mathbf{c}, \mathbf{p}, \mathbf{q})$

Alignment table T (rows = rep atoms, columns = $g \in G$):

Phase 3 segments (1):
[**0:576**] **kind=SIMPLICIAL method=simplicial_c0_deg1 len=576**: [0:576] kind `SIMPLICIAL` method `simplicial_c0_deg1` length 576

Encoded output: 576 reals, range $[-2.0000e+00,\ +4.2409e+01]$, $\max|\cdot| = +4.2409e+01$.
First values:
: +1.4142, +1.4142, +1.4142, +1.0000, +1.0000, +0.0000, +0.0000, +0.0000

Table 1: Phase 3 input alignment table T

Atom	g_0	g_1	g_2	g_3	g_4	g_5	g_6	g_7	g_8	
$\lvert\mathbf{a}\rvert$	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4142	+1.4142	+1.4142	+1.7321	+1.7
$\lvert\mathbf{b}\rvert$	+1.4142	+1.4142	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7321	+1.7
$\lvert\mathbf{c}\rvert$	+1.7321	+1.7321	+1.4142	+1.4142	+1.7321	+1.7321	+1.7321	+1.7321	+1.4142	+1.4
$\lvert\mathbf{p}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\lvert\mathbf{q}\rvert$	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0000	+1.0
$\mathbf{a}\cdot\mathbf{b}$	+2.0000	+2.0000	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+1.0000	+1.0
$\mathbf{a}\cdot\mathbf{c}$	+1.0000	+1.0000	+2.0000	+2.0000	+0.0000	+0.0000	+2.0000	+2.0000	+0.0000	+0.0
$\mathbf{b}\cdot\mathbf{c}$	+0.0000	+0.0000	+0.0000	+0.0000	+1.0000	+1.0000	+1.0000	+1.0000	+2.0000	+2.0
$\mathbf{a}\cdot\mathbf{p}$	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0
$\mathbf{a}\cdot\mathbf{q}$	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0
$\mathbf{b}\cdot\mathbf{p}$	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0
$\mathbf{b}\cdot\mathbf{q}$	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0
$\mathbf{c}\cdot\mathbf{p}$	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0
$\mathbf{c}\cdot\mathbf{q}$	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0000	+1.0000	-1.0
$\mathbf{p}\cdot\mathbf{q}$	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{c})$	-2.0000	-2.0000	+2.0000	+2.0000	+2.0000	+2.0000	-2.0000	-2.0000	-2.0000	-2.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{p})$	+1.0000	-1.0000	-2.0000	+2.0000	-1.0000	+1.0000	-1.0000	+1.0000	+2.0000	-2.0
$\varepsilon(\mathbf{a},\mathbf{b},\mathbf{q})$	-1.0000	+1.0000	+2.0000	-2.0000	+1.0000	-1.0000	+1.0000	-1.0000	-2.0000	+2.0
$\varepsilon(\mathbf{a},\mathbf{c},\mathbf{p})$	-2.0000	+2.0000	+1.0000	-1.0000	-1.0000	+1.0000	-1.0000	+1.0000	+1.0000	-1.0
$\varepsilon(\mathbf{a},\mathbf{c},\mathbf{q})$	+2.0000	-2.0000	-1.0000	+1.0000	+1.0000	-1.0000	+1.0000	-1.0000	-1.0000	+1.0
$\varepsilon(\mathbf{b},\mathbf{c},\mathbf{p})$	-1.0000	+1.0000	+1.0000	-1.0000	-2.0000	+2.0000	+2.0000	-2.0000	+1.0000	-1.0
$\varepsilon(\mathbf{b},\mathbf{c},\mathbf{q})$	+1.0000	-1.0000	-1.0000	+1.0000	+2.0000	-2.0000	-2.0000	+2.0000	-1.0000	+1.0
$\varepsilon(\mathbf{a},\mathbf{p},\mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0
$\varepsilon(\mathbf{b},\mathbf{p},\mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0
$\varepsilon(\mathbf{c},\mathbf{p},\mathbf{q})$	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0000	+0.0000	-0.0

Last 8 values:
: +24.0873, +22.4473, +36.3909, +21.8250, +9.6919, +27.9637, +20.5164, +33.6573

4 Summary

Phase 3 encoded length: Phase 3 encoded length: 576 reals

Total encoded length: Total encoded length: 576 reals

repS size: |repS| = 8 atoms

—G—: $|G| = 12$ group elements

TeX output written to: /Users/lester/github/Echinocoder/symcoder/demo_simplex2.tex Compile with: pdflatex demo_simplex2.tex